

**THE THIRTY-NINTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

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Problem A-1

Let A be any set of 20 distinct integers chosen from the arithmetic progression 1, 4, 7, ..., 100. Prove that there must be two distinct integers in A whose sum is 104.

Problem A-2

Let $a, b, p_1, p_2, \dots, p_n$ be real numbers with $a \neq b$. Define $f(x) = (p_1 - x)(p_2 - x)(p_3 - x) \cdots (p_n - x)$. Show that

$$\det \begin{pmatrix} p_1 & a & a & a & \cdots & a & a \\ b & p_2 & a & a & \cdots & a & a \\ b & b & p_3 & a & \cdots & a & a \\ b & b & b & p_4 & \cdots & a & a \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b & b & b & b & \cdots & p_{n-1} & a \\ b & b & b & b & \cdots & b & p_n \end{pmatrix} = \frac{bf(a) - af(b)}{b - a}.$$

Problem A-3

Let $p(x) = 2 + 4x + 3x^2 + 5x^3 + 3x^4 + 4x^5 + 2x^6$. For k with $0 < k < 5$, define

$$I_k = \int_0^\infty \frac{x^k}{p(x)} dx.$$

For which k is I_k smallest?

Problem A-4

A "bypass" operation on a set S is a mapping from $S \times S$ to S with the property

$$B(B(w, x), B(y, z)) = B(w, z) \quad \text{for all } w, x, y, z \text{ in } S.$$

- (a) Prove that $B(a, b) = c$ implies $B(c, c) = c$ when B is a bypass.
- (b) Prove that $B(a, b) = c$ implies $B(a, x) = B(c, x)$ for all x in S when B is a bypass.
- (c) Construct a table for a bypass operation B on a finite set S with the following three properties:
 - (i) $B(x, x) = x$ for all x in S .
 - (ii) There exist d and e in S with $B(d, e) = d \neq e$.
 - (iii) There exist f and g in S with $B(f, g) \neq f$.

Problem A-5

Let $0 < x_i < \pi$ for $i = 1, 2, \dots, n$ and set

$$x = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Prove that

$$\prod_{i=1}^n \frac{\sin x_i}{x_i} < \left(\frac{\sin x}{x} \right)^n.$$

Problem A-6

Let n distinct points in the plane be given. Prove that fewer than $2n^{3/2}$ pairs of them are unit distance apart.

Problem B-1

Find the area of a convex octagon that is inscribed in a circle and has four consecutive sides of length 3 units and the remaining four sides of length 2 units. Give the answer in the form $r + s\sqrt{t}$ with r , s , and t positive integers.

Problem B-2

Express

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{m^2n + mn^2 + 2mn}$$

as a rational number.

Problem B-3

The sequence $\{Q_n(x)\}$ of polynomials is defined by

$$Q_1(x) = 1 + x, \quad Q_2(x) = 1 + 2x,$$

and, for $m \geq 1$, by

$$Q_{2m+1}(x) = Q_{2m}(x) + (m+1)xQ_{2m-1}(x),$$

$$Q_{2m+2}(x) = Q_{2m+1}(x) + (m+1)xQ_{2m}(x).$$

Let x_n be the largest real solution of $Q_n(x) = 0$. Prove that $\{x_n\}$ is an increasing sequence and that $\lim_{n \rightarrow \infty} x_n = 0$.

Problem B-4

Prove that for every real number N , the equation

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4$$

has a solution for which x_1, x_2, x_3, x_4 are all integers larger than N .

Problem B-5

Find the largest A for which there exists a polynomial

$$P(x) = Ax^4 + Bx^3 + Cx^2 + Dx + E,$$

with real coefficients, which satisfies

$$0 < P(x) < 1 \quad \text{for} \quad -1 < x < 1.$$

Problem B-6

Let p and n be positive integers. Suppose that the numbers $c_{h,k}$ ($h = 1, 2, \dots, n; k = 1, 2, \dots, ph$) satisfy $0 < c_{h,k} < 1$. Prove that

$$\left(\sum \frac{c_{h,k}}{h} \right)^2 < 2p \sum c_{h,k},$$

where each summation is over all admissible ordered pairs (h, k) .